

Project Proposal Form

Institute of Information Technology (IIT)

University of Dhaka

Student's Name:	Mohammed Yasin		
Student's Roll:	1406	Phone:	01628898377

An AI-Driven Bangla Web Accessibility Assistant for Visually Impaired Users

Motivation

A visually impaired Bangla-speaking user often faces major difficulties while browsing modern websites using screen readers and keyboard navigation. Many webpages contain unlabeled buttons, inaccessible forms, missing image descriptions, dynamically changing content, and complex navigation structures that are difficult to interpret through assistive technologies. In most cases, accessibility information is available only in English, making interaction even more challenging for Bangla-speaking users.

As a result, users frequently struggle to understand webpage content, navigate efficiently, complete online forms, or access important notifications and updates. Existing accessibility tools mainly focus on generic or English-based support, while contextual Bangla accessibility assistance remains very limited.

This project is motivated by the need to build an intelligent accessibility enhancement system that improves webpage accessibility, generates Bangla accessibility support, enhances keyboard navigation, and provides AI-assisted webpage understanding for visually impaired Bangla-speaking users.

Project Description

This is a browser extension for Google Chrome running on the Microsoft Windows operating system. The users are required to use the NVDA screen reader as a prerequisite. The tool is an AI-driven web accessibility assistant designed to improve web interaction for Bangla-speaking visually impaired users. The system dynamically analyzes webpages and enhances accessibility by generating Bangla accessibility labels, improving semantic structures, providing accessible navigation support, translating dynamic content, and generating descriptions for visual elements. It also offers AI-assisted webpage understanding, accessibility evaluation, and personalized interaction support to create a more inclusive and accessible browsing experience.

Proposed Features

- **Context-Aware Bangla Semantic Localization:** Automatically generates meaningful Bangla accessibility labels for webpage elements using AI-assisted translation.
- **Intelligent Semantic Role Reconstruction:** Repairs non-semantic interactive webpage elements and converts them into screen-reader-friendly accessible components.
- **Adaptive Bangla Form Validation Layer:** Provides accessible Bangla validation messages and feedback during form interactions.
- **Intelligent Visual Content Interpretation:** Generates Bangla descriptions for unlabeled images using OCR and multimodal AI captioning.
- **Smart Accessibility Navigation Layer:** Improves keyboard-based and voice input navigation through intelligent skip links and landmark shortcuts.
- **Adaptive Accessibility Noise Suppression:** Reduces accessibility interruptions caused by advertisements and repetitive non-essential webpage regions.
- **Agentic Page Understanding Assistant:** Provides AI-generated Bangla webpage summaries and guided navigation assistance. Analyzes webpage accessibility quality and generates Bangla accessibility feedback for users.
- **Personalized Accessibility Adaptation Engine:** Learns user interaction patterns and dynamically adapts accessibility behavior based on user preferences.

Key Technologies & Tools

Frontend & Web Accessibility: JavaScript, HTML, CSS

Extension Development: Chrome Extension APIs, Mutation Observer API

AI & Content Processing: Google Gemini API, Tesseract OCR, Whisper

Backend: Node.js, Express.js

Database: PostgreSQL

Accessibility Testing Tools: NVDA Screen Reader

Development Tool: VS Code

Version control: Git, Github